

CS310 Project Specification

Mobile Application Enhancing Transparency of UK Politics

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1 Problem Breakdown

UK law, legislation and politics is difficult to understand. It is not a controversial statement to say that understanding the complexities and nuances of UK law and politics is no easy task. Reading any recently passed laws, you would struggle to easily decipher the meaning of the text. They are long, complex and time consuming to read. This is to the point that for any average member of the public, there is little incentive to see what MPs are actually achieving, rather just passively listening to proclamations and declarations from politicians - which poses risk of being full of half truths and misinformation. Hence, this has resulted in a decrease in trust between the UK populous and the government. Statistics released by the UK census for 2022 [4] revealed that only 35% of the population trusts the national government, and more interestingly that 58% of people are not confident people like them have a say in what the UK government does, otherwise known as poor political efficacy. These statistics alone indicate there is a gap between the public and the government that needs to be broached. Both the OECD [7], and Grotlüschen's [2] data linked literacy levels to both trust in the government and political efficacy, with Grotlüschen's finding 60% of people with a literacy level of 1 assume poor political efficacy. This is a stark difference to only 30% of those with a higher literacy level assuming poor political efficacy. This makes a strong case that at least part of the reason that trust in the government is reduced is because a portion of the population cannot easily understand the more complex reading of UK law. Thus, it can be argued that we can start to mend the relationship by making the results of what MPs do, to be more widely accessible: by making legislation easier to understand.

There are two core problems facing the comprehension of laws and bills. The first issue is with respect to the language used. Reading an in progress bill, or an amended law, is difficult due to the number of references unknown to the reader. The text itself is too complex for the average reader. This cannot be easily proven as legislative texts cannot easily be assigned a reading level in any traditional sense, Looking at the PIAAC [6], they defined six levels of literacy proficiency levels, and reading their descriptions and examples, legislation would best fall into Level 4 or 5. From a 2012 study the PIAAC found the average level amongst UK adults was Level 2 [6]. Furthermore, 16.4% of adults in England have literacy levels below the National Literacy Trust's [8] Level 1, which is considered 'very poor literacy skills'. When also including that 8.9% of the population of England does not speak English as their first language [5], it paints a picture that a sizable portion of the population would not be able to grasp UK legislation. The second issue we face is the difficulty in remaining up to date with the legislative process. Bills are regularly introduced in the house of lords or commons, to either propose a new law, or to amend an existing one. The government does provide a website where you can access a database of every bill that has been proposed, and its progress [1]. However it is difficult to keep track of bills relevant to you, The solution to both of these problems is a mobile app that updates users regularly on bills and legislation that may affect them, and provides an interface to read a simplified or summarized version of the bill - ideally simplified to the average reading level of the country. This app aims to rectify the relationship between the public and the legislative process, by making it so ordinary people can understand what bills are being passed, what laws entail in a simpler form, and letting users stay up to date on what is relevant to them.

This previously would have been impossible to do, with no relevant existing technologies that would allow for such a project. However, improvements within natural language processing have made a substantial development, and shows new promise in both the realm of text simplification, and sentiment analysis. Text simplification has five key methods behind it, seq2Seq modelling [10], edit-based sentence simplification, lexical simplification, back translation, and structural sentence simplification [3]. Each of these are used in hybrid with each other, and have several models built for each of them. The best ones will need to be selected and trained for legislative texts. There is also certain evaluation metrics/training methods we will need to look at; a mathematical way to discern the complexity of the text, as certain methods prove unusable with legislation. We will also need to explore if the text produced may have any bias; which shall use sentiment analysis models to identify [9].

2 Objectives

The project has the overarching objective of providing a method for the simplification of UK legislation. This is the greatest priority and aim for the project, not only because it is the most technical and interesting aspect, but also because no application like it exists. Framing this as the most essential part of the project, sets up the rest of our objectives, and how we are to conduct the development process.

Breaking down the problem, developing the app divides into seven key areas to consider.

Key:

- (M) = Must do
- (S) = Should do
- (C) = Could do
- (E) = Extension

2.1 Module 1: Text simplification

A module that handles the simplification process. This module will have two major facets: bills and acts. When bills are being passed through the houses, they go through a series of phases, amending each step of the way until it is accepted and passed as an act. This leads to the fact that the final writing of bills and acts have different formats, and hence must each be dealt with accordingly. While the focus of the project is to ensure that any bill can be translated into something simpler, it is likewise important that acts and laws can be simplified as well. This is because some bills are amendments of existing laws, and because we need to demonstrate what the bill looks like once it has been passed.

Requirements:

1. The module can take in a text representation of a bill or an act, pass it into a text simplification model and return a simplified form of the bill or act. (M)
2. The text result from the model is accurate to the information inside of the text. (M)
3. The text result from the model can describe a series of impacts affecting certain demographics. (C)

2.2 Module 2: Political sentiment analysis

A module to evaluate the political sentiment of the translated text. This addresses the ethical risks discussed in the problem background. We will want to find a way to interpret what the political leaning of the text may be, and if it poses any potential risk of presenting bias. This module is less of a priority, as we need the previous module to be complete before this can be worked on.

Requirements:

1. A way for the system to identify if the given text simplification is appropriate to upload. (C)
2. A method for the system to analyze the political leanings of the given text using the left-right political spectrum. (C)
3. A method for the system to analyze the political leanings of the given text using the Eysenck political spectrum. (E)
4. A method for the system to evaluate if a given text is critical or favorable toward the focus subject matter. (C)

2.3 Module 3: Live updating system

A live tracking and update system. We will want to have a way to keep track of what bills are currently being passed through the houses, their current stage, their sponsor, and any other details relevant to that bill. This information can be accessed through the UK parliament website, however a retrieval, update and storage method will need to be designed.

Requirements:

1. A system in place to download and retrieve updated bill data, including short and long title, stage, sponsor, relevant dates, bill pdf. (S)
2. A method to format the downloaded data, into something efficient to store, and feed into the other modules. (S)
3. A system in place that adds to the simplified description of the given bill, expressing what has changed from the previous stage. (C)
4. A way of notifying relevant users to the update in data. (S)
5. An assurance the app is kept up to date and accurate to current information. (S)

2.4 Module 4: Demographic identification

A method to identify what legislative texts impact what demographics. We will need to identify a series of demographics that is relevant to both the users and to the texts. This likely will be an extension of the first module, utilizing the model to find these demographics in its answer.

Requirements:

1. A method to identify what demographics are impacted by an introduced bill or an act. This demographic will be pulled from a collection of predefined demographics. (C)
2. Research into all relevant/important demographics that need to be addressed and covered by the model. (S)

2.5 Module 5: Database

An SQL database that can let us store user data, and information about the bills and legislation. The user data would be entered through self identification, designating details such as age, gender, occupation, tax bracket, etcetera. This would then need to be somehow securely stored, so as to not violate GDPR. The database storing information on the bills and legislation meanwhile will want to contain the simplified version of the legislative text, so that it is consistent in meaning and quality when distributed amongst the users. Additionally we will want to store any relevant information to the text that we would want it to display. The reason the database will be SQL is because we will want to make several complex queries surrounding the users impacted by certain demographics, and the bills that individual users may be tracking.

Requirements:

1. A well structured normalized database which contains all relevant details to be used by all other modules. (S)
2. Storing self-identified user data. This includes personal details such as: age, occupation, gender etcetera. (S)
3. Ensuring security for the storage of given user data. (M)
4. Storing important relevant information about a given bill or act. Storing the simplified text rather than the original document. (S)
5. Designed such that it can be expanded to introduce new relevant details. (C)
6. Assurances that user data will be securely stored, and will not violate GDPR. (M)

2.6 Module 6: UI

An engaging and accessible UI for users on the phone app. This project is with the aim of making legislation more accessible to users, and part of that is making it not boring. Reading legislative texts is incredibly boring, and this product is aimed at alleviating that, hence if the presentation of the UI and app is more drab than the text, it will fail the project's primary objective. The interface must be both interactive, simple and engaging, taking many attributes from modern social media apps.

Requirements:

1. Mobile app user interface designed, such that it is functional, and all elements of the interface act as expected. (S)
2. A method to ensure to write the text data from the database about the legislation, without any display errors. (S)
3. A way to access the original bill from inside of the app. (E)
4. Easy to use, engaging and interactive for the users. (S)

2.7 Module 7: Ethical discussion

Reaching a conclusion on the place of AI technologies in law and politics. Throughout this project, especially with regards to the legislative text simplification and political sentiment analysis, there will be much research put into how to design the algorithms and models so that there will be no inaccuracy or bias. However, there is no guarantee of success for either front. This project is using its given product as a medium to investigate the future of this technology, and concluding if it's ever moral or ethical to use technology in this way.

Requirements:

1. Research the use of technology, AI, and automation in the realm of law and politics. (S)
2. Evaluate if it is ethical to use these technologies in law and politics. (S)
3. Evaluate if it can ever be safe to use these technologies in law and politics. (S)
4. Evaluate if AI technologies can ever truly be unbiased in politics. (S)
5. Use evidence of existing case studies, and use findings from the results of this project's model and technologies. (S)

2.8 Other Extensions:

1. An ability for the user to leave comments on a given bill or act simplification. (E)
2. An ability for users to vote/request new simplifications on given bills or acts. (E)
3. Additional information and resources on the legislation page produced by third party sources, such as ground news. (E)
4. An educational opening which informs the user in simpler terms how the legislative process actually works. (E)
5. Make the app more accessible to a larger number of people, with accessibility settings such as text size, audio prompts and readings, and in-built definitions for words which may not be as easily understood by all users. (E)

3 Methodology

The methodology used will be Agile. Agile allows us to bring the most important part of the project, and that is the flexibility. Due to how much research and experimentation that this project involves, it is incredibly likely that a goal or an approach will have to change to accommodate this, especially since certain modules are highly dependent on other modules working. Over the course of the next 7 months, I will be undertaking 2 week long sprints, where each sprint will focus on research, implementation, development and then testing. At first the iterations will focus on designing and improving module 1, the text simplifications but as the project develops, we will branch out the focus into the other modules accordingly. This process will involve a lot of reevaluation stages, where the we will refocus time and resources on to the respective modules that need it the most. Modules are functionally independent from each other, and theoretically each can be done on their own, which leads to what needs to be prioritised. Ordering the modules in priority of highest to lowest is: 1, 3, 5, 6, 4, 2. Modules will likely be done in parallel as the project progresses, needing to adapt to requirement amendments during reevaluation. Once the first module is functional enough for the project to progress, parallel development will begin.

4 Timetable

For the timetable, you can see in the Gantt chart below, the plan I intend to follow for term 1. This term will focus primarily on module 1, 3 and 5. This set up is so that it focuses on the priority modules for the software, and then allows term 2 to focus on the development of the UI, the demographic identification, and if there is time, the political sentiment evaluation. This has been done to avoid scope creep, as there are many unknowns with this project, and by following this timetable, we will maximise the effectiveness of what we complete.

The first two sprints (starting from week 3) focus on module 1, and aims to have it working to some extent by the end of the second sprint. This opens the door for focus on module 3 and 5. Module 6 is begun at the end of term, however with term 1 coursework for other modules being due at that time, it may not receive much work until after the term has ended, hopefully leading into after Christmas.

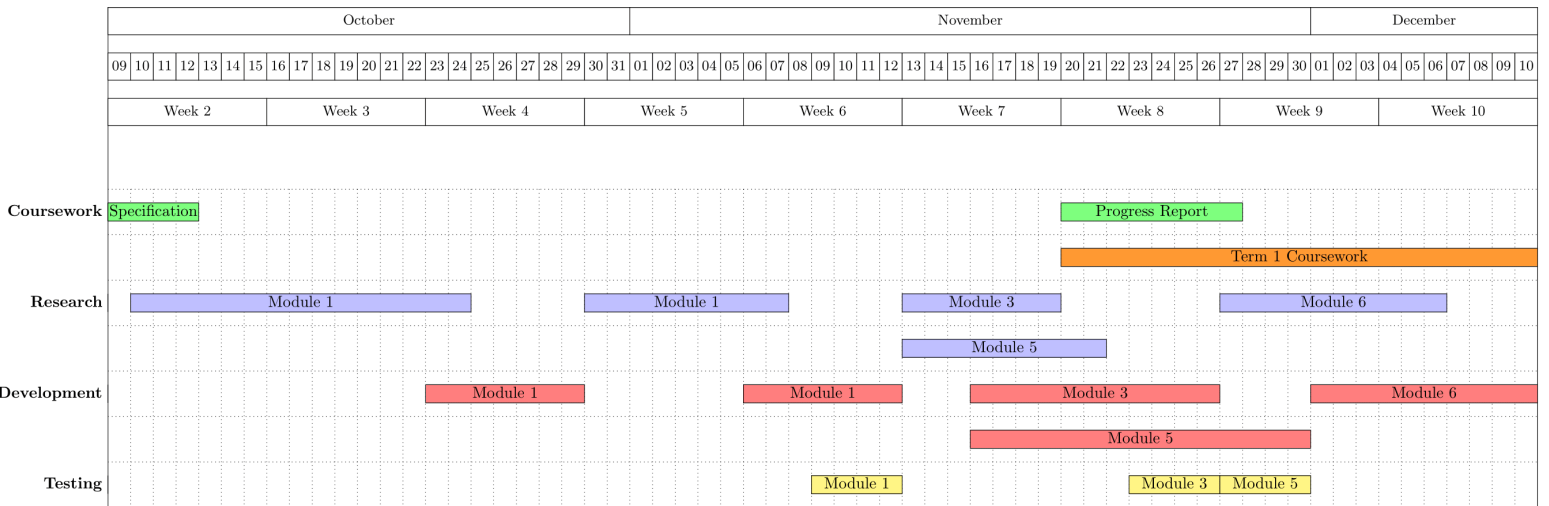


Figure 1: Term 1 Gantt Chart

5 Resources

Initial known resources for the project. May change or expand during project development.

- GitHub: This will be used for version control of the code as it is developed.
- Text simplification models: There are many existing models that may suit my purposes. These include Hugging Face, PyTorch (Seq2Seq), Tensor2Tensor, EditNTS, LSBert, Glavas, and LaserTagger.
- Text datasets: These will be useful in simplification, however available ones I have found may not best account for legislative dialogue and will need expansion. These datasets may still find some use though: X-Sum, D-Wikipedia and Newsela.
- Political sentiment models: There are less models that look at this, but two which show promise are PyABSA, and Textblob. This will definitely be expanded upon as the project develops.
- UK parliament website: This will be used to access every legislative document used.
- Mobile app development tool: An app development tool which will be used for development; will need to be able to make calls to other modules, and the database.
- SQL database server: Some method of implementing a server to upload to and query from across devices, likely using AWS or Azure.

6 Risks

Here are the given risks for the project, and how their impact will be reduced or mitigated:

- Other courseworks or deadlines getting in the way:
Ensuring there is enough redundancy in my time management.
- Difficulty learning mobile app development:
Ensuring there is enough redundancy in my time management. Worst case scenario switching to web development if upon evaluation appears too difficult.
- Government website goes down and cannot access their files:
Download an appropriate subset of files at random that can be used for use and testing.
- Lack of storage space for all legislation document details:
Research into the use of external storage software, such as cloud, or through the department.
- Development PC suffers damage such that the files on it are inaccessible:
GitHub will be regularly updated with the project and its resources.

- Bias within the system. There is a real risk that the model or training data used, could accidentally skew the simplifications to certain political affiliations. This risks radicalising people, which can lead to increased extremist actions, or even inspire criminal activity against the government:
Attempt to implement a political sentiment analysis tool, to prevent this from having extreme bias. It will not be released to the public until some assurance of no extreme bias. This will need an independent review during the testing process by myself to ensure the text produced is suitable.

7 Legal, Professional and Ethical Considerations

There are a couple legal considerations, and importantly ethical issues that need to be discussed.

Legal issues:

- GDPR concerns: I need to make sure that any personal data stored by a user is securely stored inside of the database. Alternatively, all data provided by the user is fictitious. This will depend on the development of the project and will be reevaluated.
- Defamation risks: This is a risk of information spread by the app is incorrect. This inherently doesn't pose any legal risks, however during testing it will be important to inform the testers that information may not be accurate. Hopefully as development progresses, methods to avoid this risk will be researched and found.

Ethical issues:

The risks of accuracy and bias, within the system and outside of it, leads to an uncomfortable ethical question with regards to our recent advances in technology. Should we use automated AI systems, within the realm of law and politics? The issue arises from bias, as there is evidence that AI can be trained on bias data, which can result in the model itself being biased with dangerous impacts on groups and individuals. This product may influence the populous negatively with a biased take on the legislation. Furthermore, if this product was released publicly, maintained and run by a person, business or organization, then this tool could be used to intentionally skew public opinion on certain bills and laws. This would inadvertently turn the product into a tool for propaganda. This feeds into the question of if it is possible to have an unbiased system, especially with something so subjective as politics. This project will aim to explore what it means to use AI, and to a large scope, technology, inside of politics; questioning if it is possible to create something that is consistently unbiased.

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